TIME SERIES SYNTHETIC DATA GENERATION

Intorduction to data security

Work Presented by:

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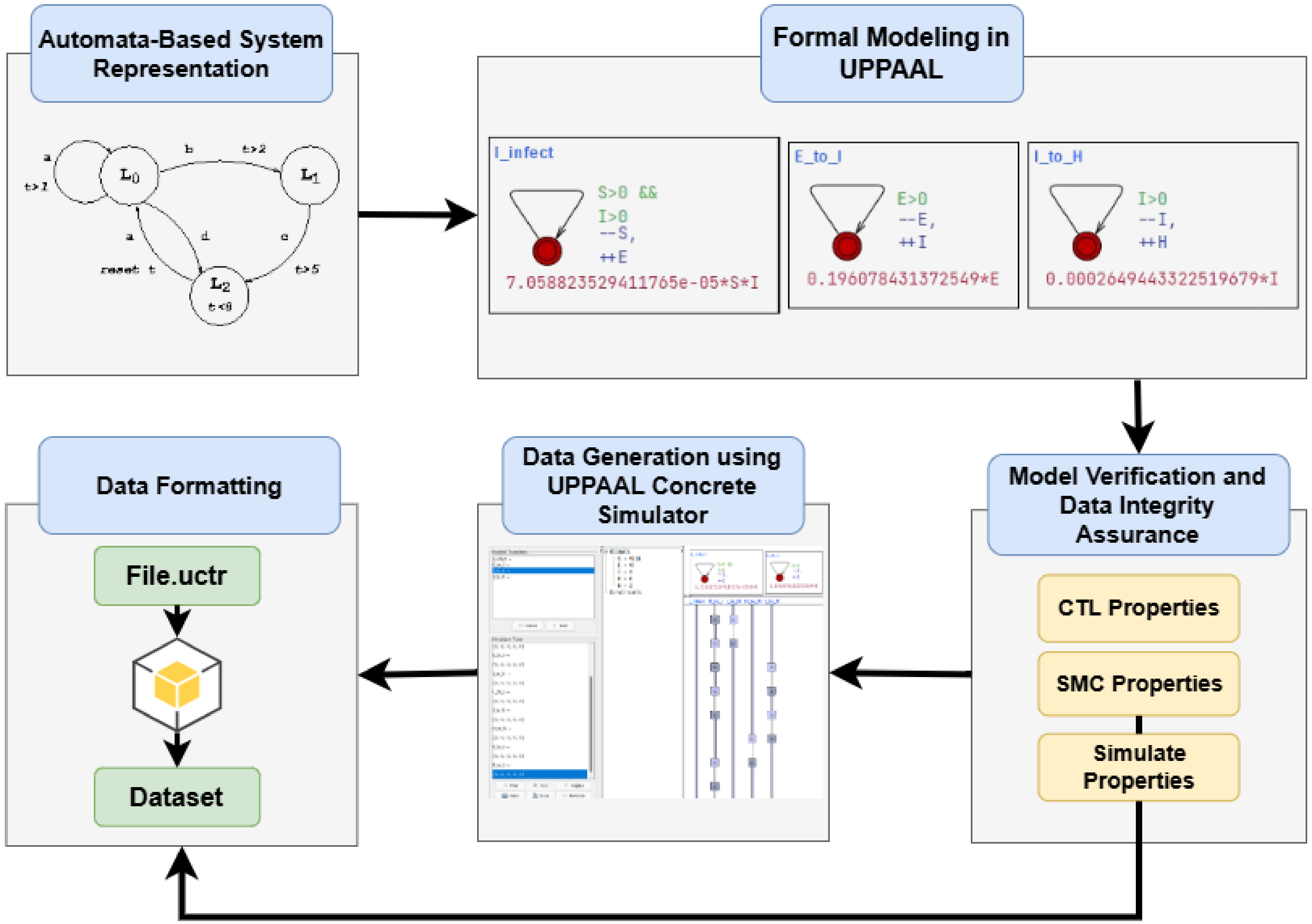
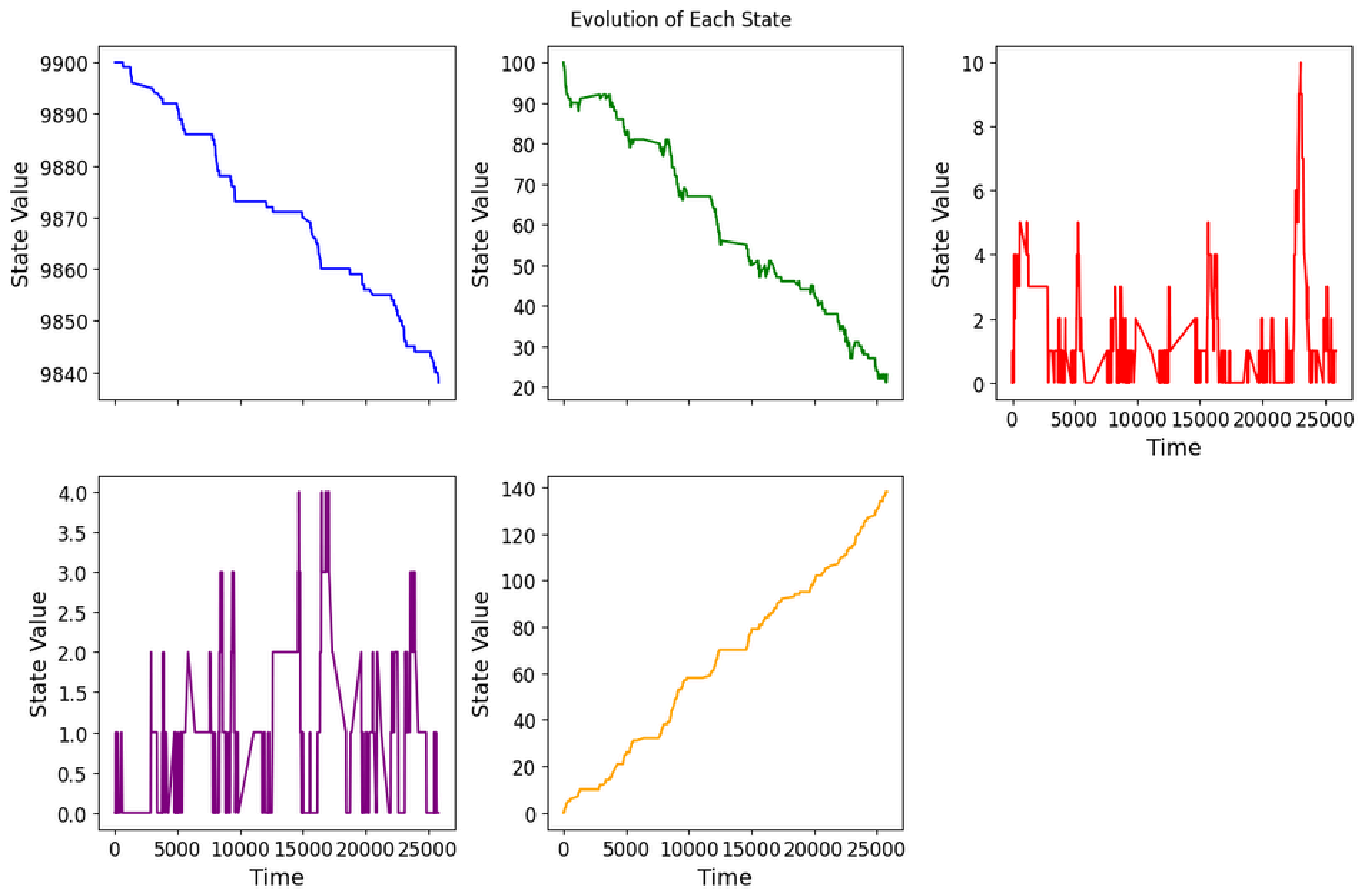
The need for time series synthetic data generation

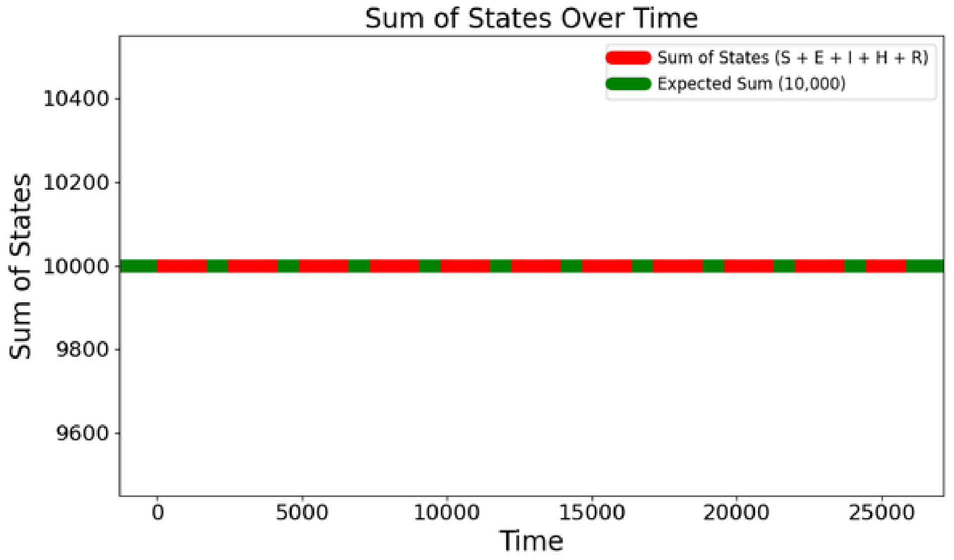
1. **Lack of real data**
2. **Privacy and security:** Sharing or using real time series data might breach privacy laws (e.g., GDPR, HIPAA). Synthetic data can mimic real data without exposing confidential information.
3. **Data augmentation:** In situations where real time series datasets are small or imbalanced, synthetic data can augment them to improve model performance.
4. **Simulating rare events**: Events like financial crashes, equipment failures, or cybersecurity breaches are rare. Synthetic data allows us to simulate such critical cases and prepare models to handle them.
5. **Cost and feasibility:** Collecting real-world time series can be time-consuming and expensive.

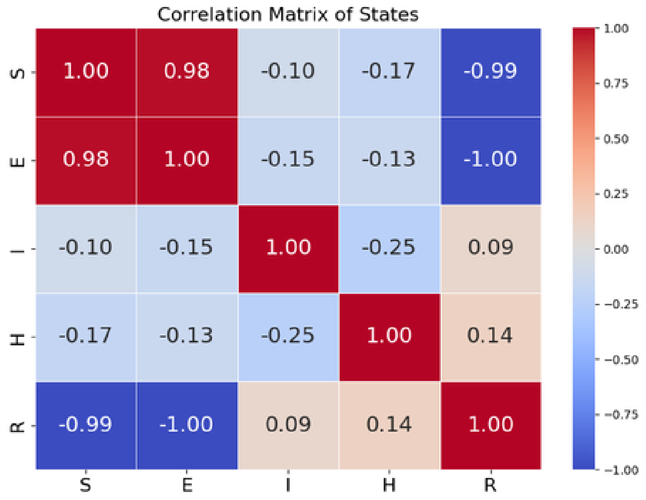
Github Link : https://github.com/ilyesrezgui/Intro_Data_Security/tree/ilyes-branch

Formal methods for time series synthetic data generation: A model-checking approach









Tools used: UPPAAL, P!YTHON, XML,, MATPLOTLIB, SKLEARN, DRAW.IO

Work by Ilyes REZGUI **Data generated over 25000 time stamps, and for 10000 person in a population**

Privacy-Preserving Synthetic Time Series Data Generation Using DP-GAN and DP-TimeGAN

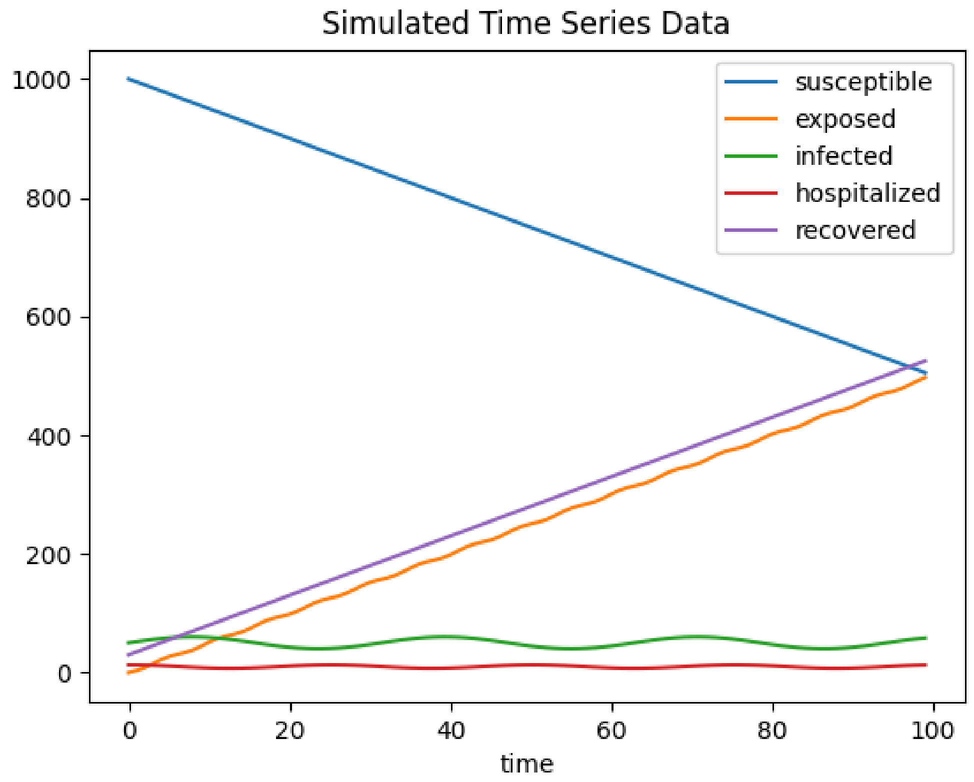
Goal:
- Generate synthetic time series data while preserving privacy using Differential Privacy (DP) techniques.

Models Implemented:
- DP-GAN: Standard GAN with Differentially Private SGD (DP-SGD) applied to the Discriminator.
- DP-TimeGAN: Modified TimeGAN with privacy-preserving Discriminator using Opacus.

Key Techniques:
- Differential Privacy enforced via Opacus.
- Privacy Parameters: $\varepsilon = 10.0$, $\delta = 1e\text{-}5$.
- Discriminator trained with DP; Generator remains non-privat

Dataset:
- Synthetic SEIHR epidemic data
- 100 time steps × 5 features
- Normalized using MinMaxScaler

Results:
- Generated synthetic sequences closely mimic real temporal patterns.
- Privacy constraints respected with acceptable ε.

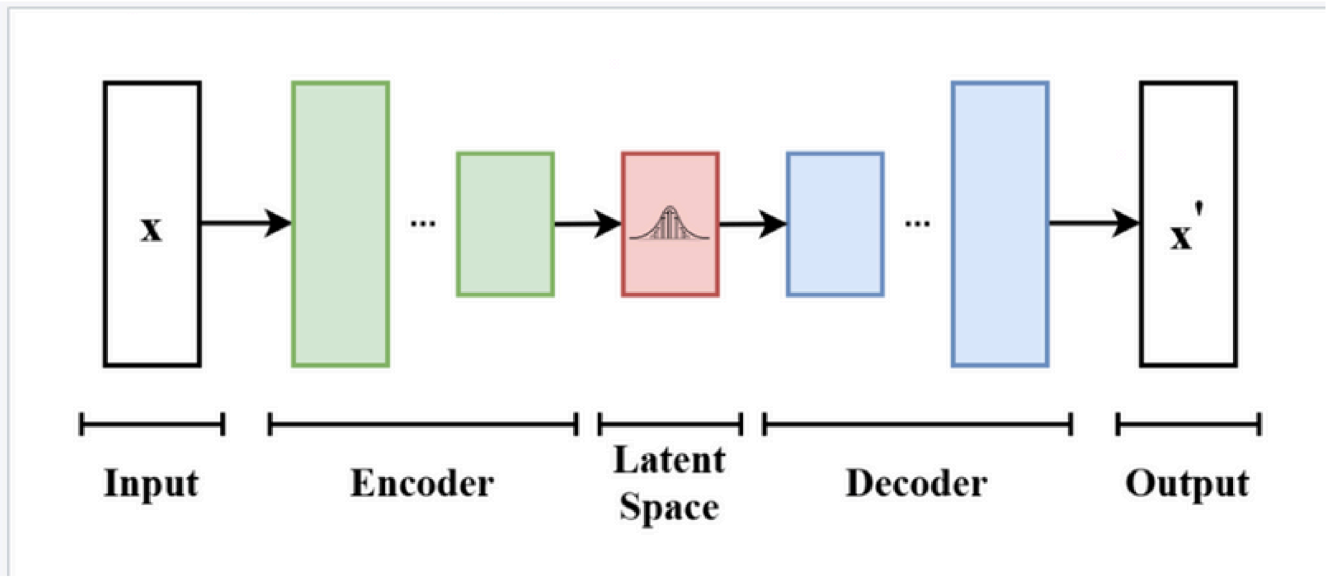


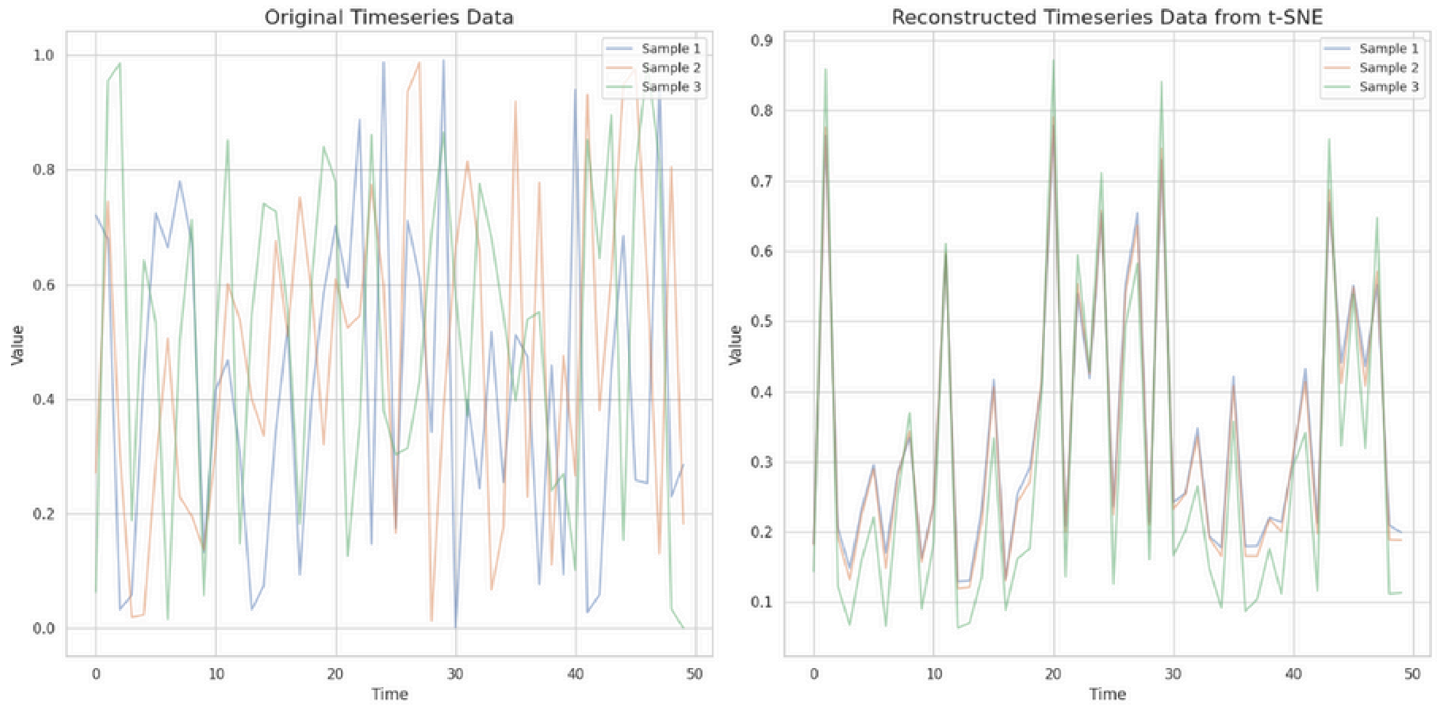
Work by Kanan ORUJOV

Variational Autoencoder (VAE) for data generation

Description:

- This project implements a Variational Autoencoder (VAE) using TensorFlow.
- The model learns patterns from synthetic timeseries data and reconstructs it through a latent space.
- Training involves a custom loop with a combination of Reconstruction Loss (MSE) and KL Divergence to regularize the latent space.
- After training, the reconstructed timeseries closely matches the original.
- Main goal: Visualize VAE performance on synthetic timeseries.





Work by Ines MAHJOUB

TimeGAN for synthetic data generation

- **Data Preparation:**

I created a synthetic dataset consisting of 50,000 samples and 4 features. The data was normalized between 0 and 1 using MinMaxScaler, and organized into sequences of 24 time steps to preserve temporal dependencies.

- **Model Design:**

I built a generator model based on Long Short-Term Memory (LSTM) networks. The model learned to generate new time series sequences by minimizing the Mean Squared Error (MSE) loss.

- **Training and Generation:**

The model was trained using the Adam optimizer for 10 epochs. After training, synthetic sequences were generated by feeding random noise into the model.

Visual comparisons between real and synthetic sequences showed that the generator successfully mimicked real-world time series patterns.

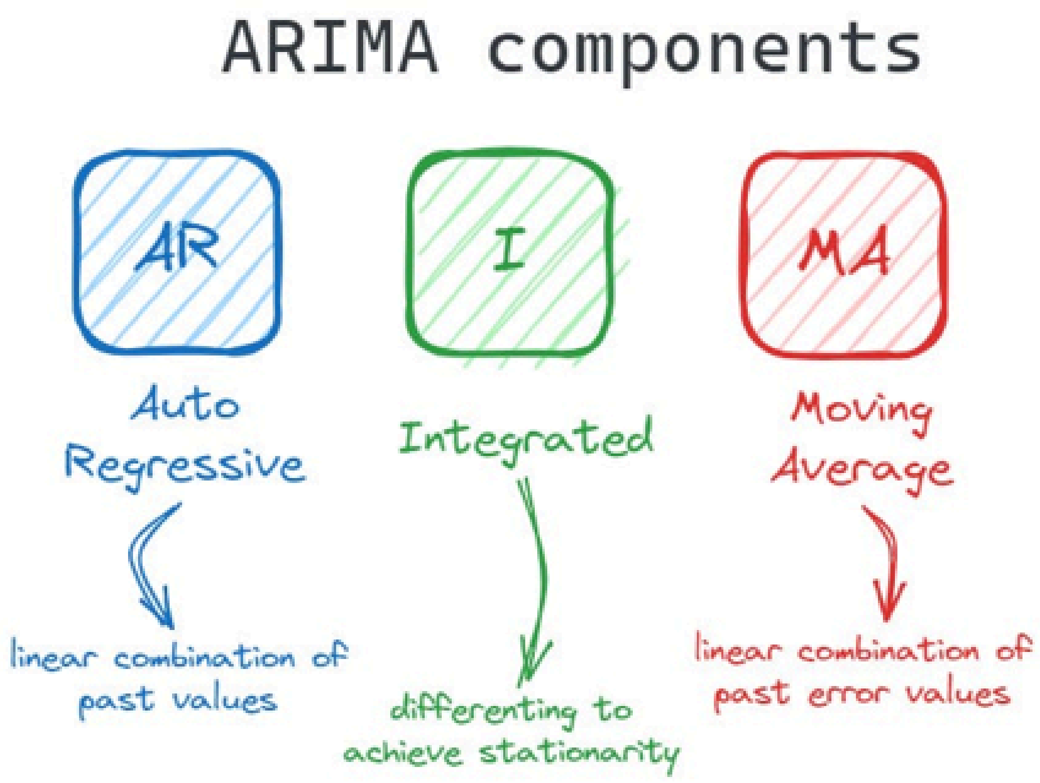
- **Model Inversion Concept:**

I explored the idea of model inversion techniques, where the goal is to reconstruct underlying data distributions by analyzing the trained model's behavior.

Tools used

Work by Abdullazade ROYAL

Autoregressive Integrated Moving Average Model

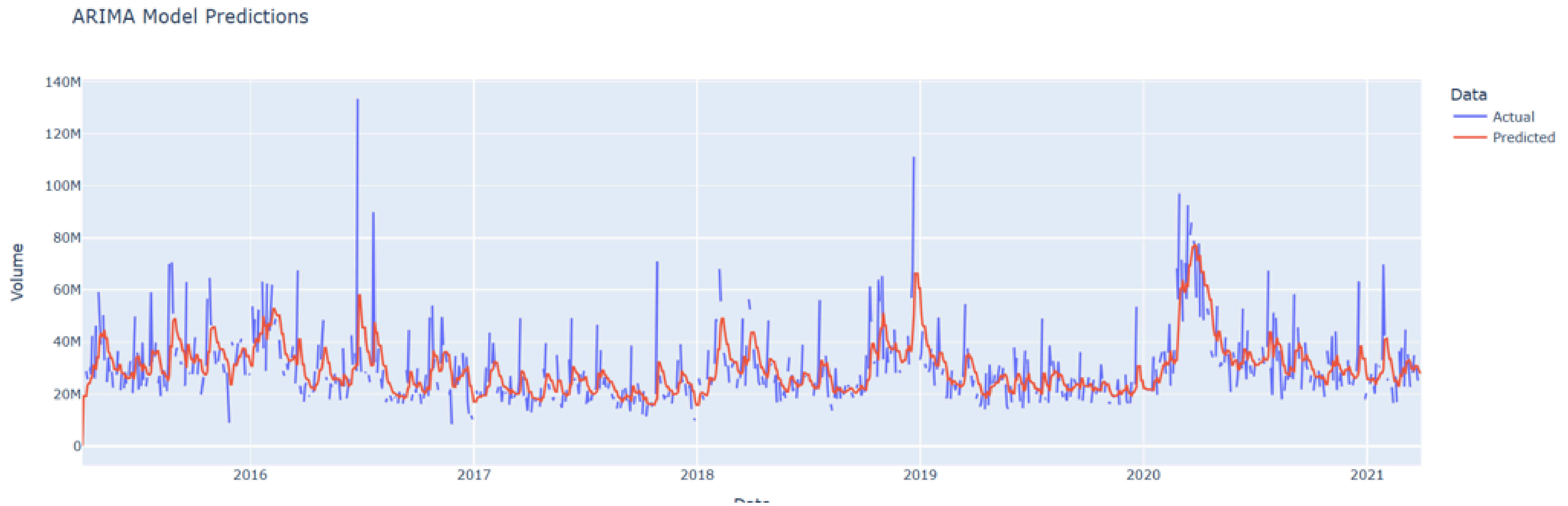


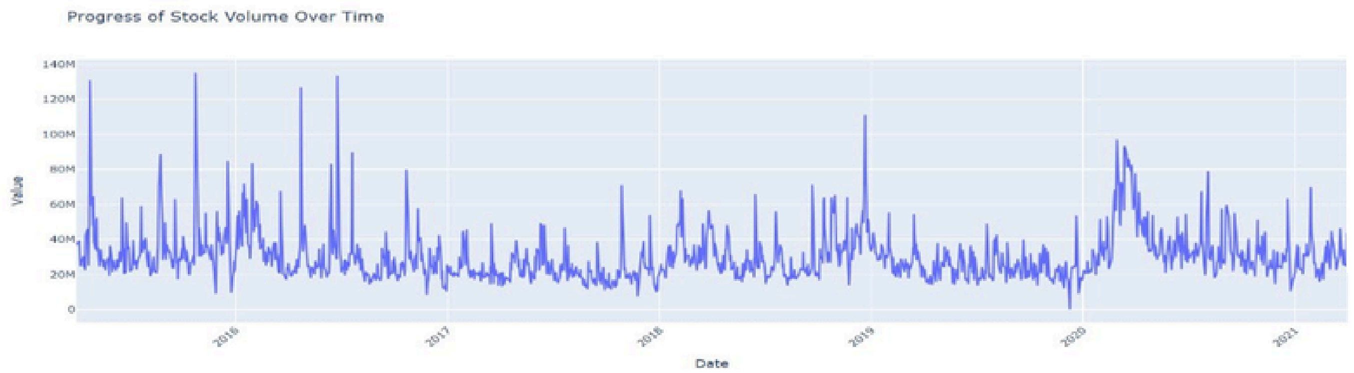
$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + .. + \beta_p Y_{t-p} + \epsilon_1$$

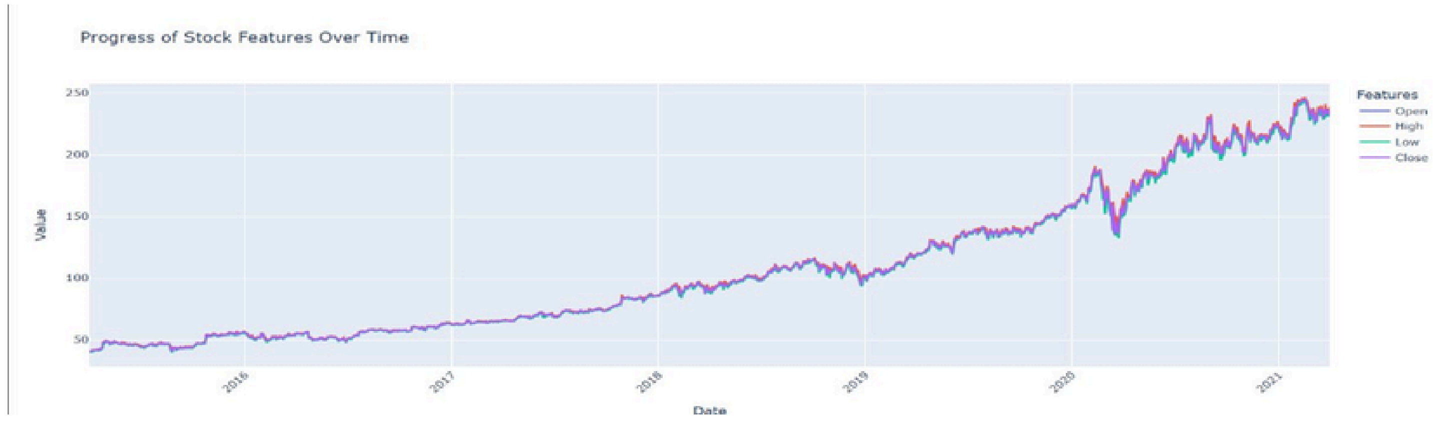
$$Y_t = \alpha + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + .. + \phi_q \epsilon_{t-q}$$

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + .. + \beta_p Y_{t-p} \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + .. + \phi_q \epsilon_{t-q}$$

ARIMA







Work by Fares GHEZAL

Dataset: Microsoft Stock- Time Series Analysis

Conclusion

Formal methods for time series synthetic data generation: No need for a reference dataset, data is generated from a formal model, A complete new approach that was never studied in the literature, It is scalable and available to use, it includes a formal verification process which assures the correctness of the data.

Differentially Private Generative Models for Time Series:
This project used DP-SGD with DP-GAN and TimeGAN to generate realistic, privacy-preserving synthetic time series data. It balances utility and privacy, is reproducible with Opacus and PyTorch, and supports privacy-aware AI practices.

TimeGAN for synthetic data generation: This project shows that a simple LSTM-based generator, inspired by TimeGAN, can effectively learn and reproduce time series patterns. The approach is effective because it combines simplicity with strong performance in capturing temporal dependencies.
Future work can explore adversarial training and model inversion to improve data fidelity.

ARIMA: This project demonstrates that the ARIMA model can effectively generate realistic synthetic time series data. Its statistical foundation and interpretability make it a practical choice for applications requiring simple, transparent, and privacy-conscious data generation.

Variational Autoencoder (VAE):
This project used VAEs to learn compact and informative latent representations of time series data, enabling the generation of realistic synthetic sequences. The model captures complex temporal dependencies, supports unsupervised learning, and offers flexibility for various downstream tasks. It is suitable for scalable, privacy-preserving data generation without requiring strict statistical assumptions.

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THANK YOU

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